

General Psychology

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1 Chapter 1: The Story of Psychology

1.1 1.1 Defining the Discipline

Psychological Science is the scientific study of the **mind** (subjective experiences), **brain** (biological mechanisms), and **behavior** (observable actions).

Psychology differs from common sense by grounding its conclusions in **empirical investigation**, not intuition or cultural sayings.

Psychiatry vs. Psychology:

- **Psychiatry:** Medical field (MD), can prescribe medications, treats biological causes of disorders.
- **Psychology:** Research, therapy, behavioral interventions. Mostly non-medical, except in limited jurisdictions.

1.2 1.2 Origins and Big Debates

Psychology began in philosophy and branched into science through two core debates:

Nature–Nurture Debate: Genes provide the biological base, but environment shapes expression. “Nurture works on what nature endows.”

Mind–Body Problem: Is the mind separate from the body, or does the mind emerge from brain activity? Modern cognitive neuroscience supports the latter.

1.3 1.3 Early Schools of Thought

Structuralism (Wundt, Titchener): Used introspection to identify the structure of mental processes. Failed due to subjectivity.

Functionalism (William James): Influenced by Darwin; focused on adaptive functions of behavior and consciousness. Practical, experience-driven.

Behaviorism (Watson, Skinner): Rejected mental processes; studied only observable behavior. Dominant in early 20th century.

Cognitive Revolution (1960s): Shift back to studying thoughts, memory, and internal mental processes.

1.4 1.4 Modern Psych Perspectives (The 7 Lenses)

- **Biological / Physiological:** Brain mechanisms, hormones, genetics.
- **Cognitive:** Thinking, memory, problem solving, interpretation.

- **Behavioral:** Effects of reinforcement, punishment, conditioning.
- **Psychodynamic:** Unconscious conflicts, early childhood experience.
- **Evolutionary:** Adaptive traits shaped by natural selection.
- **Developmental:** Lifespan changes in behavior and cognition.
- **Socio-Cultural:** Influence of society, culture, norms, and context.

1.5 1.5 The Scientific Method

The scientific method ensures objectivity:

1. **Develop a hypothesis:** Must be operationalized and falsifiable.
2. **Controlled test:** Experimental group (treatment) vs. control group.
3. **Gather objective data.**
4. **Analyze:** Accept or reject hypothesis.
5. **Publish and Replicate.**

1.6 1.6 Major Research Methods

- **Experiment:** Only method establishing cause-effect.
- **Correlation:** Shows association, not causation.
- **Naturalistic Observation:** Behavior in natural contexts.
- **Survey:** Large samples; subject to wording bias.
- **Case Study:** Deep analysis of rare or unique individuals.

1.7 1.7 Classic Example: Zimbardo's Prison Study

Demonstrated the “Lucifer Effect”—situations can override personality and push normal individuals toward harmful behaviors. Highlights the power of context and social roles.

2 Chapter 2: Attention and the Two-Track Mind

2.1 2.1 Dual Processing: Two Minds in One

The brain operates on two tracks:

- **High Road (Conscious Sequential Processing):** Slow, effortful, deliberate. Used when learning or solving new problems.
- **Low Road (Unconscious Parallel Processing):** Fast, automatic, handles routine tasks. Expert driver vs. novice driver example.

Key demonstration: Writing a complex sentence while counting backward. Both require conscious processing → interference.

2.2 2.2 Selective Attention

Humans intake ~11 million bits per second but consciously process about 40 bits.

Selective Attention = cognitive spotlight. Everything outside the spotlight is filtered out.

2.3 2.3 Selective Inattention

Inattentional Blindness: Failure to notice visible objects (gorilla experiment).

Change Blindness: Failure to detect changes in environment (door-swap experiment).

2.4 2.4 Major Theories of Attention

2.4.1 A. Spotlight Theory

Attention works like a movable beam; whatever falls inside is processed deeper.

2.4.2 B. Early Selection Theory (Cherry, 1953)

Dichotic listening showed people ignore unattended message content → filtering based on **physical characteristics**, not meaning.

2.4.3 C. Late Selection Theory

Meaningful information (e.g., your name) can penetrate the filter. “Cocktail Party Effect.”

2.4.4 D. Feature Integration Theory

Explains how features combine:

- **Parallel Search:** Unique feature pops out.
- **Serial Search:** Target shares features with distractors → requires focused attention to bind features.

2.4.5 E. Biased Competition Theory

Neural inputs compete for resources; salient or intense inputs dominate (shattered glass example).

2.5 2.5 Biological Basis: Blindsight

Damage to visual cortex leads to conscious blindness but preserved ability to navigate or grasp objects → two visual pathways:

- **Perception track:** Conscious recognition.
- **Action track:** Unconscious visuomotor guidance.

2.6 2.6 Multitasking

Humans cannot perform two conscious tasks simultaneously. We “task-switch,” causing attentional costs and increased errors (e.g., texting while driving).

3 Chapter 3: Sensation and Perception

3.1 3.1 Sensation vs. Perception

Sensation: The process by which sensory receptors and the nervous system detect and transmit stimulus energies from the environment. **Perception:** The organization and interpretation of sensory input into meaningful objects and events.

Example: Sensation = detecting vibrations. Perception = interpreting it as a “doorbell sound.”

3.2 3.2 The Mystery of Perception

Although retinal images constantly shift as we move, we perceive the world as stable. This requires complex neural computation involving:

- **Feature detection**
- **Parallel processing**
- **Top-down & bottom-up integration**
- **Perceptual constancies**

3.3 3.3 How Sensation Becomes Perception

3.3.1 Bottom-Up Processing

Stimulus-driven; begins at receptors and builds upward. Necessary for unfamiliar or ambiguous stimuli.

3.3.2 Top-Down Processing

Uses prior knowledge, expectations, and context. Drives perceptual set and quick interpretation.

3.3.3 Feature Detectors

Discovered by **Hubel & Wiesel**: Neurons in the occipital lobe respond selectively to edges, angles, shape, or movement.

Face-specific regions: Right temporal lobe (fusiform face area) — damage causes prosopagnosia (cannot recognize faces).

3.3.4 The Binding Problem

How does the brain combine color, shape, orientation, and motion into a unified perception? This remains an open neuroscientific challenge.

3.4 3.4 Parallel Processing

The brain processes motion, form, depth, and color **simultaneously**. This enables extremely fast perceptual recognition.

Example from textbook neuroscience diagrams (pages around 6.20–6.22): Face recognition recruits distributed networks; 30% of the cortex is visual processing.

3.5 3.5 Perceptual Constancy

The ability to recognize objects as unchanging despite variations in lighting, distance, or angle.

3.5.1 Color Constancy

Objects appear the same color even under changing illumination.

Checker-Shadow Illusion: A square in shadow appears lighter than identical square outside shadow (explained by brightness constancy + context).

3.5.2 Shape Constancy

We identify objects as having constant shapes despite changing retinal images. Example: Opening door → trapezoid image → still perceived as a rectangle.

3.5.3 Size Constancy

Objects are perceived as constant in size despite distance. Explains why people appear “smaller” at a distance but are known to be the same size.

3.6 3.6 Illusions: When Perception Fails

Illusions arise when interpretation mechanisms misfire.

3.6.1 Conditions that enhance illusions

- Unclear or ambiguous stimuli
- Missing information
- Conflicting depth cues
- Cultural differences (context-specific learning)

3.6.2 Classic Illusions

1. Hermann Grid Grey dots at intersections due to lateral inhibition (retinal ganglion cells).

2. Vase–Faces Illusion Ambiguous figure-ground switching; alternation between hemispheric dominance.

3. Checkerboard Illusion Brightness and color constancy mechanisms override raw sensory data.

4. **Müller-Lyer Illusion** Effect of “carpentered world” (Gregory, 1977): Zulus in circular culture were *less* susceptible because their environmental context lacked right angles.
5. **Ponzo Illusion (added)** Linear perspective makes identical lines appear different lengths.
6. **Ames Room (added)** Manipulated room geometry creates distorted size perception.

3.7 3.7 Gestalt Principles of Perceptual Organization

Gestalt psychology argues that the mind is innately wired to perceive patterns and wholes.

Core idea: The whole is greater than the sum of its parts.

3.7.1 Figure–Ground

We separate objects (figures) from background (ground). Occurs automatically and continuously.

3.7.2 Grouping Laws

- **Similarity:** Similar elements group together.
- **Proximity:** Nearby elements group together.
- **Continuity:** Smooth, continuous patterns preferred.
- **Closure:** We fill gaps to perceive whole objects.

3.7.3 Law of Prägnanz

The brain prefers the simplest, most stable interpretation requiring the least mental effort.

3.8 3.8 Depth Perception

3.8.1 Evidence from Infant Studies

Gibson & Walk (1960): Visual Cliff Infants older than 6 months avoid the drop → depth perception is partly innate.

Bower (1971): Virtual 3D Stimuli Heart rate increased when infants viewed objects approaching → early depth sensitivity.

3.8.2 Binocular Cues

Require both eyes.

Retinal Disparity: Greater disparity = closer object. Forms basis for stereoscopic vision.

Convergence (added): Eyes inward-turning for close objects.

3.8.3 Monocular Cues

Work with one eye.

- **Motion Parallax:** Nearby objects move faster across the retina.
- **Atmospheric Perspective:** Distant objects look hazier/bluer.
- **Linear Perspective:** Parallel lines converge at distance.
- **Relative Size:** Smaller retinal image → farther.
- **Interposition:** One object blocking another → closer.
- **Texture Gradient (added):** Closer objects have more detailed textures.

3.9 3.9 Perceptual Set and the Role of Learning

Perception is shaped by experience, expectations, and context.

Perceptual Set: Mental predisposition to perceive one thing and not another.

Examples:

- “D?CK” → DUCK after seeing animals; → DICK after seeing names.
- Cultural grouping:
 - Western: Category-based (cow + chicken = animals)
 - East Asian: Relation-based (cow + grass = cow eats grass)

3.9.1 Context-Specific vs. Context-Independent Perception

Socio-cultural factors determine whether perception emphasizes:

- **Figure** (object-focused)
- **Ground** (context-focused)

3.10 3.10 Summary

This chapter establishes:

- Sensation provides raw data.
- Perception interprets these data using neural, cognitive, and cultural systems.
- Our perceptions are not perfect reflections of reality—they are intelligent guesses.

4 Chapter 4: Learning

4.1 4.1 What is Learning?

Learning: A lasting change in behavior or mental processes resulting from experience.

Learning vs. Instinct Instincts are innate, species-typical behaviors (e.g., spider spinning a web). Learning is acquired, flexible, and shaped by environment.

4.2 4.2 Classical Conditioning (Pavlov)

Focuses on learning an association between two stimuli (S–S learning). The response is **involuntary** and reflex-based.

4.2.1 4.2.1 Pavlov’s Discovery

Over 532 experiments on “psychic secretions”—dogs salivating to footsteps. The reflex was being triggered by a previously neutral stimulus.

4.2.2 4.2.2 Core Components

- **Unconditioned Stimulus (US):** Naturally triggers response (food).
- **Unconditioned Response (UR):** Reflexive reaction (salivation).
- **Conditioned Stimulus (CS):** Previously neutral (tone).
- **Conditioned Response (CR):** Learned response to CS.

Example: Reading a menu and salivating (top-down + conditioned association).

4.2.3 4.2.3 Key Processes

Acquisition: Initial stage where CS–US pairing strengthens CR.

Extinction: CR disappears when CS is repeatedly presented without US.

Spontaneous Recovery: Reappearance of weakened CR after rest period.

Generalization: Similar stimuli provoke CR (bell vs buzzer).

Discrimination: Distinguishing between CS and similar stimuli. *Pavlov’s dogs* learned ellipse = shock, circle = food.

4.2.4 4.2.4 Applications: Little Albert

Watson & Rayner (1920): Conditioned fear in infant Albert by pairing loud noise with a white rat. Generalization occurred: fear of rabbits, fur coats, Santa beard.

Major takeaway: Emotions can be classically conditioned.

4.3 4.3 Operant Conditioning (Skinner)

Behavior is shaped by its **consequences**. Focuses on voluntary responses (R–S learning).

4.3.1 4.3.1 The Law of Effect (Thorndike)

Behaviors followed by satisfying outcomes are strengthened; behaviors followed by unpleasant outcomes are weakened.

4.3.2 4.3.2 Reinforcement

Reinforcement **strengthens** behavior.

Positive Reinforcement: Adding a desirable stimulus. Example: Salary at end of the month.

Negative Reinforcement: Removing an aversive stimulus. Example: Wearing warm clothes to reduce cold.

Important: Negative reinforcement is **NOT** punishment.

4.3.3 4.3.3 Types of Reinforcement

Continuous Reinforcement: Every response is reinforced. Useful for shaping complex behavior but produces rapid extinction.

Intermittent Reinforcement: Responses reinforced sometimes. Produces strongest and most resistant behavior.

4.3.4 4.3.4 Schedules of Reinforcement

- **Fixed Ratio (FR):** Reinforce after set # of responses. E.g., free pizza after 10 purchases.
- **Variable Ratio (VR):** Reinforce after unpredictable # of responses. E.g., Slot machines — highest response rate.
- **Fixed Interval (FI):** Reinforce after fixed time. E.g., Checking the oven near timer end.
- **Variable Interval (VI):** Reinforce after varying time. E.g., Checking for messages/likes on Instagram.

4.3.5 4.3.5 Punishment

Punishment **weakens** behavior.

Positive Punishment: Adding aversive stimulus (speeding ticket).

Negative Punishment: Removing desired stimulus (taking away phone).

Is punishment effective? Short-term behavior suppression, long-term ineffectiveness.
Can create fear, avoidance, aggression.

4.4 4.4 The Four Big Challenges to Behaviorism

Behaviorism claimed all learning is conditioning. These four landmark findings proved otherwise.

4.4.1 4.4.1 Tolman: Latent Learning

Tolman's maze experiments: Rats explored maze without reinforcement yet learned layout. When food introduced, they instantly took shortest route.

Conclusion: They formed a **cognitive map**. Learning can occur without reinforcement.

4.4.2 4.4.2 Köhler: Insight Learning

Chimpanzees solved problems suddenly (stacking boxes to reach bananas). Not trial-and-error → "Aha!" moment.

Conclusion: Learning involves mental restructuring, not just conditioning.

4.4.3 4.4.3 Seligman: Learned Helplessness

Dogs exposed to unavoidable shocks later failed to escape even when escape was possible.

Key idea: Internal belief of *no control* suppresses behavior.

4.4.4 4.4.4 Biological Limits on Conditioning

Garcia & Koelling (1966): Taste Aversion Rats easily associate taste with nausea—NOT with shock or noise. **Preparedness:** Evolution favors certain associations.

Keller & Breland (1961): Instinctive Drift Trained animals revert to biological predispositions. (e.g., raccoons "washing" coins instead of depositing them)

4.5 4.5 Observational Learning (Bandura)

Learning occurs by watching others.

4.5.1 Bobo Doll Study

Children imitated aggressive acts seen in adults. Strongest predictor: **what they see**, not what they're told.

4.5.2 Mirror Neurons

Neurons that fire when performing an action *or* watching another perform it. Biological basis for imitation and empathy.

4.6 4.6 Why Habits Are Hard to Break

Habits form due to:

- Repeated classical conditioning (coffee + sweet foods)
- Strong operant reinforcement
- Intermittent reinforcement schedules (most resistant to extinction)

Breaking habits requires:

- Removing reinforcement
- Introducing replacement behaviors
- Conscious cognitive control (top-down)

4.7 4.7 The Premack Principle

A preferred activity can reinforce a less-preferred one. “You can play video games after finishing your notes.”

4.8 4.8 Summary Table: Conditioning vs. Cognition

Topic	Behaviorism Claim	Modern Finding
Classical Conditioning	Any CS can pair with any US	Biological preparedness restricts pairing
Operant Conditioning	Behavior shaped only by consequences	Internal expectations matter (Tolman)
Learning Mechanisms	Trial-and-error, reinforcement	Insight, cognitive maps, active problem solving
Emotion Learning	Purely environmental	Biological + cognitive + social factors